

SMARTFOOD: An LLM-Driven Ubiquitous Mobile App for Context-Aware Restaurant Recommendation

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Abstract

We introduce SMARTFOOD, a context-aware mobile app and rich user profiles (dietary restrictions, allergies, taste preferences, health status) to deliver personalized, restaurant-level food recommendations that promote healthier choices. By combining nutritional insights from the Finnish Fineli database with a fine-tuned LLM, SMARTFOOD not only suggests venues and dishes that fit each user's unique context—whether a quick office lunch or a family dinner—but also highlights options lower in calories. In performance tests on three Android configurations, app startup time ranged from 2.2 s to 3.4 s, lunch-menu retrieval from 6.5 s to 7.5 s, and screen loads under 0.8 s. Average memory usage was 370 MB (emulators) to 470 MB (S22 Ultra) with peak CPU under 55%, indicating robust cross-device performance. In a small unmoderated user study (N=5), all participants achieved 100% task completion (account setup, menu fetching/feedback, profile updates) with low error rates (<4.1 %) and mean times per task under 3.7 minutes. Usability scales were high (SUS = 83.6 ± 8 , UMUX-LITE = 80 ± 6), confirming the app's ease of use and effectiveness. Our source code is available at <https://github.com/ValtteriJokisaari/smartfood>.

CCS Concepts

• **Applied computing** → **Health informatics**; • **Human-centered computing** → **Ubiquitous computing**.

Keywords

Large Language Models (LLM), Restaurant Recommendation, Ubiquitous Computing, Mobile Applications

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1 Introduction

Obesity has reached epidemic proportions worldwide, presenting severe health and economic burdens [1, 20]. In 2022, over one billion adults were classified as living with obesity, leading to healthcare costs approximately 30% higher than those for individuals with normal BMI [3, 10]. In Finland, nearly one million adults over 30 carry at least 15kg of excess weight, and childhood overweight rates stand at 29% for boys and 18% for girls [15]. This condition elevates the risk of type 2 diabetes, several cancers, and depression, underscoring the urgent need for effective preventive strategies.

Dining outside the home which is common for many, poses heightened risks for individuals with allergies or intolerances, which affect 15–20% of the population, due to fragmented or ambiguous menu information [19]. Consumers also demand concise nutritional summaries to make informed, swift choices, yet most restaurant- and diet-focused apps emphasize ratings, cuisine type, or location over personalized health guidance [2].

AI has emerged as a transformative force in health promotion and dietary management. By analyzing vast datasets and user behavior, AI-driven tools can deliver real-time, tailored feedback and recommendations that support sustained behavior change [18]. The undeniable role of AI in shaping people's dietary habits is evident in personalized nutrition platforms that adapt meal plans based on health profiles, preferences, and lifestyle factors [7, 17].

Despite these advances, no existing tool fully integrates individual characteristics (e.g., allergies, health goals, taste preferences), contextual data (e.g., location, time of day, activity level), and comprehensive nutritional databases to recommend healthier dining options at the point of decision.

To address this gap, we introduce **SMARTFOOD**, a cross-platform mobile prototype that leverages on-device sensors, fine-tuned large language models, and the Finnish Fineli open nutrition database. SMARTFOOD generates personalized, context-aware restaurant and dish recommendations with explicit focus on healthy choices, prioritizing lower calories, saturated fats, and added sugars, tailored to each user's unique profile. A continuous feedback loop of user



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ratings and usage logs further refines the recommendation engine over time, improving both accuracy and user satisfaction.

This work demonstrates the feasibility of combining context-aware inference, fine-grained nutritional estimates, and LLM-driven personalization in a unified system. We detail our design requirements, prototype implementation, and preliminary evaluation results, positioning SMARTFOOD as a foundation for future research on personalized, health-promoting food recommendation systems.

2 Food/Restaurant Recommendation Apps

Mobile dining apps have revolutionized how consumers discover and select restaurants by offering on-the-go access to reviews, menus, and reservations. Leading platforms such as Yelp and Zomato combine collaborative filtering with sentiment analysis to surface highly rated venues based on location and user interactions, while delivery-focused services like Uber Eats and DoorDash leverage deep learning and graph neural networks to recommend dishes optimized for immediate ordering and timely fulfillment. Other solutions, including Google Maps and TheFork, integrate natural language processing and multi-factor scoring to balance popularity, availability, and personal history. Despite their technical sophistication and high user ratings, these systems commonly prioritize convenience metrics (speed or crowd-sourced popularity) over individualized health considerations (Table 1).

Traditional recommendation strategies fall into three main categories: collaborative filtering identifies patterns in user-item interactions, content-based filtering analyzes dish or restaurant attributes, and hybrid methods blend both. Recent advances have introduced reinforcement learning and LLMs to parse free-text reviews or conversational inputs, enabling more flexible, conversational querying. Yet, these approaches still rely heavily on historical ordering behavior or aggregate ratings, which may not reflect a user’s evolving dietary objectives, allergies, or medical restrictions. Moreover, they seldom leverage real-time context—such as time of day, current activity, or immediate geographic surroundings—to refine suggestions at the moment of decision.

Beyond algorithmic limitations, user experience constraints persist. Many apps provide extensive nutritional data via static labels or menus, but these interfaces can overwhelm users seeking quick, health-focused guidance. The absence of an adaptive feedback loop means recommendations do not improve meaningfully from individual ratings or long-term health outcomes.

In contrast, SMARTFOOD addresses these gaps by uniting fine-grained nutritional insights from the Fineli database, context-aware inference from on-device sensors, and an LLM-driven personalization engine. Unlike existing platforms, SMARTFOOD explicitly incorporates user health status (e.g., BMI goals, allergies), dynamically fuses current context (location, activity), and highlights heart-healthy or low-sugar options. Crucially, its continuous feedback loop captures real-time user ratings and behavioral data to iteratively refine recommendations. This health-centric, adaptive approach not only enhances relevance for each dining occasion but also supports sustained dietary improvements—an aspect largely absent from today’s mainstream restaurant recommendation apps.

Table 1: Comparison of Restaurant Recommendation Apps

Application	Recommendation approach	Ap-Health / Context Focus
Yelp	Collaborative filtering and sentiment analysis, check-ins, and location data	No health-specific features
Zomato	Content-based and sentiment-aware filtering	No health-specific features
Google Maps	Hybrid filtering with LLM integration	Peak-time and busyness data only
OpenTable	NLP and collaborative filtering	Basic allergy filters
Uber Eats	Deep learning and graph learning	No nutritional data
DoorDash	Semantic embeddings and graph neural networks	No explicit health filtering
SMARTFOOD	Context-aware, fine-tuned LLM with nutritional database and dynamic user profiling	Explicit recommendations, adaptive to real-time context and user feedback

3 System Overview

Figure 1 provides a high-level view of SMARTFOOD’s modular architecture, illustrating how the mobile client, cloud backend, data acquisition pipelines, and AI-driven recommendation engine.

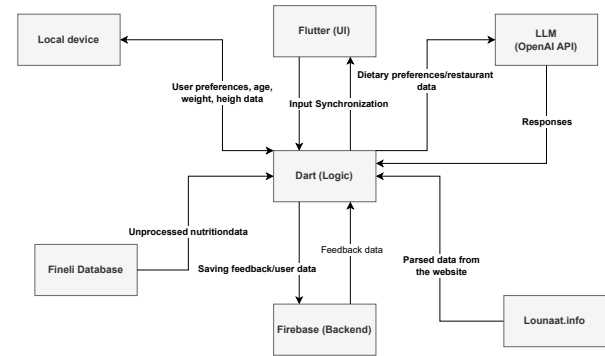


Figure 1: SMARTFOOD system architecture.

3.1 Mobile Client

The SMARTFOOD application is implemented in Flutter (Dart) to deliver a single codebase across Android, iOS, web, and desktop platforms [11, 13] (Figure 2). The client handles:

- **User Interface:** Widget-based screens for sign-in, preferences, menu browsing, and feedback, leveraging hot-reload for rapid iteration.
- **Context Sensing:** On-device collection of GPS, time-of-day, and activity status to enrich recommendation context.
- **Local Persistence:** SharedPreferences stores lightweight user state (credentials, preferences, BMI, default city) for fast startup and offline resilience [4].
- **Data Binding:** Reactive UI elements (dropdowns, sliders, buttons) reflect live recommendation results and feedback summaries.

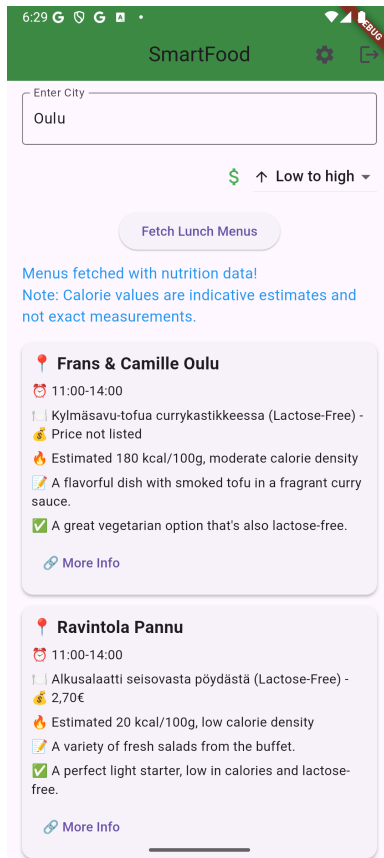


Figure 2: Menus fetched in Oulu.

3.2 Cloud Backend

We utilize Firebase for authentication, data storage, and serverless logic [12]:

- **Authentication:** Firebase Auth with Google Sign-In secures user access and provides a unique user ID.
- **Data Storage:** Firestore holds user profiles, dietary settings, scraped menu entries, and structured feedback.
- **Cloud Functions:** Serverless endpoints orchestrate menu scraping, LLM prompt construction, nutrient lookup, and feedback summarization.
- **Analytics:** Google Analytics for Firebase logs usage metrics to guide iterative improvements.

3.3 Data Acquisition

Lunch Menu Scraping. A Dart-based scraper fetches daily lunch menus from *lounaat.info* [14] which is Finland’s most popular and comprehensive service aggregating lunch menus from over 2,200 restaurants nationwide. City names are sanitized and appended to the base URL; CSS selectors (e.g., `.menu .item`) extract restaurant metadata and dish lists (name, price, dietary tags). Parsed entries are serialized into JSON objects for downstream processing.

Nutritional Database. We embed a preprocessed Fineli “Peruspaketti 2” dataset (4 232 foods, 74 nutrients) as a local SQLite asset

[5, 6]. A two-table schema (foods, energy) enables efficient lookup of energy values (kJ/100 g), converted to kcal. Data integrity is ensured via UTF-8 normalization and unit-standardization of fields.

3.4 Calorie Estimation Pipeline

The *enhanceWithNutrition* service applies:

- (1) **Normalization:** Clean dish names (lowercasing, punctuation removal, Finnish lemma approximation).
- (2) **SQL Matching:** Token-based SQL LIKE queries yield candidate food IDs.
- (3) **Similarity Ranking:** Cosine similarity on token-frequency vectors selects the best match ($\cos(\theta) > 0.4$).
- (4) **Energy Conversion:** Retrieved kJ values are converted to kcal; unmatched items defer to LLM-estimated densities.

3.5 LLM-Driven Recommendation Engine

SMARTFOOD’s core logic uses GPT-4o-mini via OpenAI API [16]. The recommendation flow is:

- (1) **Payload Assembly:** Client sends user profile (allergies, BMI, goals), context (location, time, activity), and calorie-annotated menus.
- (2) **Prompt Engineering:** A template guides the LLM to filter by restrictions, rank by health metrics (calories, fats, sugars), and generate rationale for each suggestion.
- (3) **Response Parsing:** Regular expressions extract structured fields (restaurant, dish, kcal, commentary) for rendering in the UI.

3.6 Continuous Feedback Loop

User feedback (ratings, likes/dislikes, comments) is stored in Firestore. A Cloud Function periodically:

- Aggregates historical feedback and generates a summarized profile via LLM.
- Updates prompt templates to reflect evolving user tastes and health outcomes.
- Triggers analytic dashboards to monitor recommendation efficacy.

This end-to-end design, comprising cross-platform UI, scalable backend services, robust data acquisition, precise nutrition estimation, and adaptive LLM prompts, enables SMARTFOOD to deliver real-time, personalized, and health-promoting restaurant recommendations while remaining extensible for future research.

4 Evaluation

We conducted various evaluation experiments to assess the system’s performance and verify core components, including menu scraping, feedback storage, authentication, and calorie estimation. The system reliably scraped daily restaurant menus from *lounaat.info* and stored all relevant data without loss. User feedback was successfully submitted, stored in Firestore, and reflected in local summaries. Authentication via Firebase correctly handled login, logout, and error states. Calorie estimation accurately matched single-ingredient dishes with the Fineli database, while composite dishes were reasonably estimated using heuristics and LLM fallback.

4.1 Performance Testing

We measured startup time, menu retrieval latency, screen-load times, memory footprint, and CPU utilization across three devices (Table 2). Each test was repeated 15 times per device.

Table 2: Performance metrics across test devices

Metric	Pixel (Emu)	9	Pixel (Emu)	S22 (Phys)	Ultra
App startup (s)	3.2		3.4	2.2	
Lunch menu retrieval (s)	7.3		7.5	6.5	
Screen load (s)	0.5		0.8	0.4	
Memory usage (MB)	370		355	470	
CPU utilization (peak%, idle)	40 / 2		40 / 2	55 / 15	

Overall, SMARTFOOD exhibited acceptable performance: startup and retrieval times under 8 s, memory usage below 500 MB, and CPU peaks within device capabilities. Emulators reported only application-level CPU, whereas physical-device measurements included system overhead. UI responsiveness remained smooth, with only isolated overflow warnings on unusually long strings.

4.2 User Study

We recruited five participants with no prior training. Each completed three tasks: (1) registration and preference setup, (2) menu browsing and feedback submission, (3) profile modification. We recorded task completion rate, time-on-task, and error rate (misclicks or invalid operations), then administered SUS and UMUX-LITE questionnaires [8, 9]. As can be seen in Table 3 All participants completed tasks successfully. Mean task times were under 4 min, and error rates are below 5 %. SUS and UMUX-LITE scores (> 80) indicate perceived usability. Qualitative feedback highlighted the intuitive layout and clarity of feedback controls, while suggesting the removal or redesign of the age input dropdown and further improvement of text wrapping for long dish names.

Table 3: User study quantitative results (N=5)

Task	Completion (%)	Time (s)	Error Rate (%)
Registration & preferences	100	115	2.2
Browse / feedback	100	221	4.1
Modify profile	100	184	2.8
SUS score	83.6 (± 8)		
UMUX-LITE score	80 (± 6)		

These results demonstrate that SMARTFOOD is functionally robust, performs efficiently across devices, and delivers a user-friendly experience. Identified areas for refinement, LLM formatting artefacts and UI overflow handling will be addressed in the next development to enhance system reliability and user satisfaction.

5 Conclusion and Future Work

In this work, we addressed the pressing challenge of enabling healthier dining decisions in an era of rising obesity and fragmented menu information. Our objectives were to (1) design a cross-platform mobile prototype that personalizes restaurant and dish recommendations based on user health profiles, allergies, and real-time context;

(2) integrate a reliable nutritional knowledge base for accurate calorie estimation; and (3) implement a continuous feedback loop to refine recommendations over time. To achieve these goals, we combined Flutter and Firebase for rapid front- and back-end development, a Fineli-derived nutritional database for authoritative energy values, a Dart-based web scraper for live lunch menus, and GPT-4o-mini as an LLM-driven ranking engine.

Our methodology comprised:

- **Data acquisition:** Structured scraping of menu items, pre-processing and local caching of the Fineli data, and secure storage of user profiles and feedback in Firestore.
- **Calorie estimation:** A hybrid pipeline that matches normalized dish tokens against Fineli entries (via SQL LIKE and cosine similarity), converts kJ to kcal, and defers to LLM estimates when necessary.
- **LLM-driven ranking:** Prompt engineering to fuse user context, health constraints (BMI, goals, allergies), and nutritional facts into a single template that instructs GPT-4o-mini to filter, rank, and justify healthy menu options.
- **Iterative refinement:** A serverless feedback summarization workflow that updates prompt templates and context-fusion weights based on explicit user ratings and comments.

Key novelties and performance:

- **Health-centric personalization:** Explicit optimization for low-calorie, low-fat, and low-sugar meals rather than aggregate popularity or proximity.
- **Context-aware fusion:** Real-time integration with user profile and feedbacks for more personalized recommendations.
- **Hybrid nutrition pipeline:** Combining authoritative Fineli nutrient values with LLM fallback for proprietary dishes.
- **Efficient cross-platform delivery:** Sub-8 s menu retrieval, < 500 MB memory footprint, and > 80 SUS/UMUX-LITE usability scores across heterogeneous devices.

Limitations and future directions:

- **Menu data dependency:** Reliance on `lounaat.info` may be brittle; future work will explore APIs, restaurant partnerships, and user-contributed menus to enhance coverage.
- **Calorie estimation accuracy:** While single ingredient matches are precise, multi ingredient and restaurant-specific dishes still rely on approximate LLM estimates. We plan to incorporate user-submitted portion photos and portion-size feedback to train a specialized vision-and-LLM hybrid estimator.
- **Expanded context signals:** To further personalize recommendations, we will integrate wearable or smartphone activity data so that suggested meals align with each user's current energy expenditure and fitness goals.
- **Privacy and control:** As we incorporate more personal and sensor data, we will enhance transparency and user control over data sharing, possibly adding on-device LLM inference to minimize cloud dependencies.

SMARTFOOD lays a robust foundation for research and deployment of health-promoting, context-aware food recommender systems. By iteratively refining our nutrition pipeline, broadening menu sources, and enriching personalization with wearable data, we aim to close the loop on sustainable dietary behavior change.

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